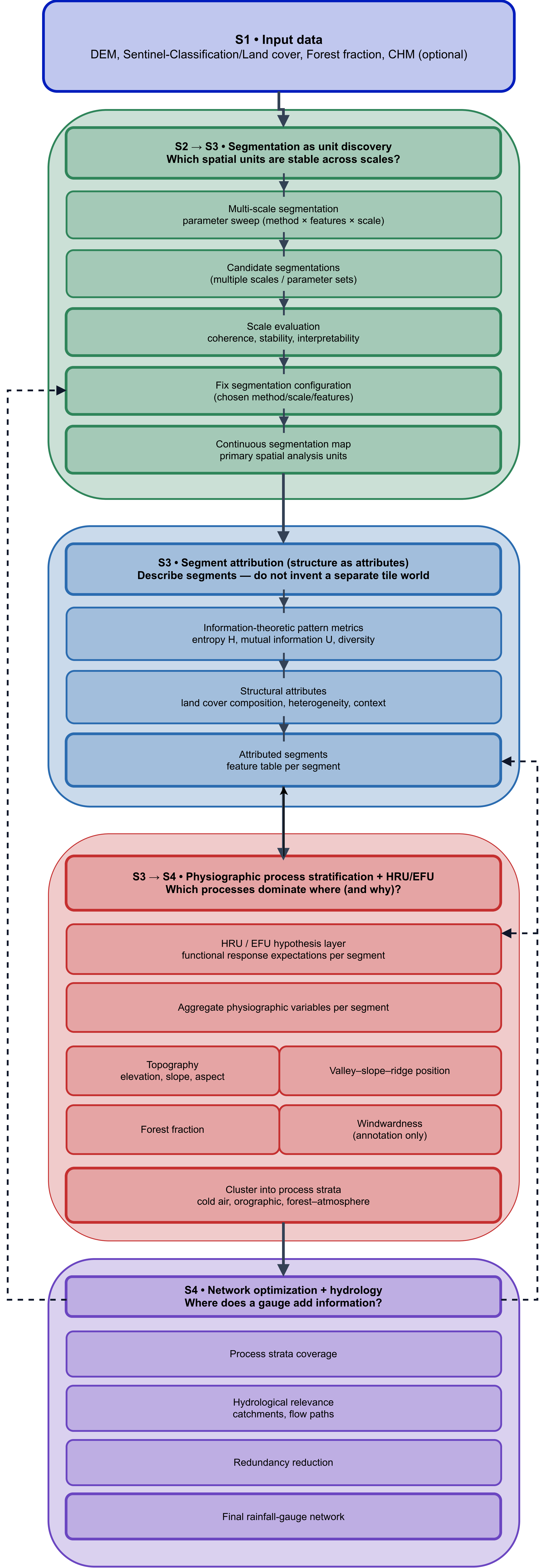
S1 • Input data
DEM, Sentinel-Classification/Land cover, Forest fraction, CHM (optional)
S2 → S3 • Segmentation as unit discovery
Which spatial units are stable across scales?
Multi-scale segmentation
parameter sweep (method × features × scale)
Candidate segmentations
(multiple scales / parameter sets)
Scale evaluation
coherence, stability, interpretability
Fix segmentation configuration
(chosen method/scale/features)
Continuous segmentation map
primary spatial analysis units
S3 • Segment attribution (structure as attributes)
Describe segments — do not invent a separate tile world
Information-theoretic pattern metrics
entropy H, mutual information U, diversity
Structural attributes
land cover composition, heterogeneity, context
Attributed segments
feature table per segment
S3 → S4 • Physiographic process stratification + HRU/EFU
Which processes dominate where (and why)?
HRU / EFU hypothesis layer
functional response expectations per segment
Aggregate physiographic variables per segment
Topography
elevation, slope, aspect
Valley–slope–ridge position
Forest fraction
Windwardness
(annotation only)
Cluster into process strata
cold air, orographic, forest–atmosphere
S4 • Network optimization + hydrology
Where does a gauge add information?
Process strata coverage
Hydrological relevance
catchments, flow paths
Redundancy reduction
Final rainfall-gauge network